

Rainfall Model

Dataset Understanding

- The training dataset contained information from 20 sites
- Data comes mainly from two sources: GOES and HRRR
 - **GOES:** sensor data
 - **HRRR:** prediction data
- GOES updated every 5 minutes, but rainfall data updates every 10 minutes.
- HRRR updated every 1 hour
- Additional data includes location and timestamp
- 1hr context is provided for each day of observation
- Rainfall greater than 1mm/hr was highly infrequent (only ~2.43% of observations)

Data Audit

- **HRRR:** Values persisted for an hour. Generally no gaps.
- **GOES:**
 - **Rainfall:** Updates every 10 minutes with missing values between updates
 - **Cloud Optical Density:** Theoretically updates every five minutes with values forward filling for the next five minutes, but MANY missing values. Some outside reading suggests that COD data is often hard to read which leads to many missing observations.
- Both sources include age counters to track how recently data was reported.
- **Modeling Consideration:** The repetition/persistence patterns of measurements made me shy away from a CNN approach. I opted instead for a standard neural net with lagged features capturing context.

Data Treatment

- **Spatiotemporal data:**

- The 'day_of_year' and 'hour_utc', 'minute_utc' are converted into sines and cosines
- Same done with 'lat' and 'lon'
- Given the minimal number of sites and days, I considered dropping them entirely, but instead I decided to let the model decide how much to weight these. Locale and seasonality seem too important to discard entirely.

- **GOES**

- 'goes_rain_rate_mm_hr' and 'goes_rain_max_50km' are forward filled.
- NaN values for COD data are filled with 0's and an associated mask column is generated which indicates whether the value was originally missing or not.

Feature Engineering

- **GOES Rainfall:**

- Lagged features of GOES rainfall data in 10, 20, and 40 minute delays
- Difference and “difference of differences” from 10 minute back

- **GOES COD:**

- Day/Night ('is_day') column added using pvlib. This is meant to differentiate whether COD data is 0 because of clear skies or nighttime.
- Lagged features at 5, 10, 25 minute delays

- **HRRR**

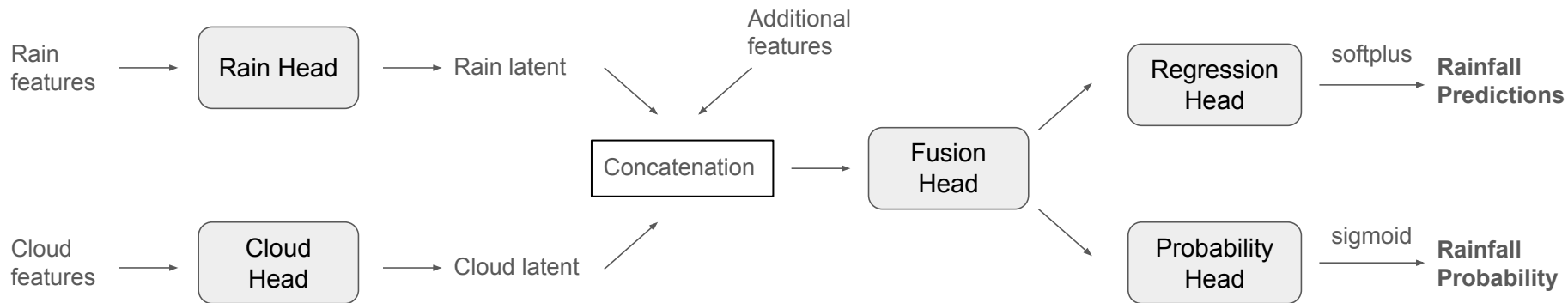
- Single lagged features 1 hr back

Note: Other features were experimented with including rolling mean, max, std.dev. along with difference and “difference of differences.” These were ultimately omitted for contributing to overfitting and/or just reducing accuracy.

Model Design

- **Multi-Head Architecture:** Rainfall and Cloud features are fed into separate processing heads embedding them in latent spaces of the same size. These are then concatenated along with other features before being fed into a final processing head.
- **Complexity:**
 - There is a notable difference in update frequency and quantity of features associated with rainfall data and cloud optical data. The dual heads allow us to maintain roughly equal importance for both rainfall and cloud signals rather than allowing cloud features to dominate.
 - GOES and HRRR are two separate sources for data and the multi-head approach allows for a unified rainfall and cloud signal.

Model Architecture



Loss Calculation and Training

- **Hybrid loss function:**
 - $\text{Loss} = 0.3 * \text{MSE_Loss} + 0.7 * \text{BCE_Loss}$
 - To emphasize the importance of accurately predicting rain events despite their low frequency, BCE_Loss was set to `pos_weight=14`.
- Three separate models are trained with identical architecture for each time horizon to be predicted. Each model was trained for 15 epochs.

Compare:

- **Baseline:** Linear Ridge Regression and Logistic Regression on HRRR data

Results

t+1	BCE	AUC - ROC	Brier Score	RMSE	MAE	RMSE (rain)	MAE (rain)
Baseline	0.394	0.923	0.116	1.403	0.211	8.881	3.565
Model	0.134	0.968	0.041	1.396	0.135	8.762	3.244

t+3	BCE	AUC - ROC	Brier Score	RMSE	MAE	RMSE (rain)	MAE (rain)
Baseline	0.397	0.921	0.117	1.403	0.211	8.885	3.567
Model	0.138	0.965	0.043	1.385	0.134	8.742	3.247

t+5	BCE	AUC - ROC	Brier Score	RMSE	MAE	RMSE (rain)	MAE (rain)
Baseline	0.400	0.920	0.118	1.403	0.211	8.885	3.567
Model	0.152	0.967	0.047	1.392	0.132	8.759	3.211

Operational Considerations

Operational Concerns:

- **Concern:** The distinction between heavy rain and very heavy rain in the mm/hr regression is likely less meaningful than improper classification of rain events to trigger rerouting. False positives and negatives probably present a greater risk to the consumer than regression error.
- **How I addressed it:** There was a tradeoff as I tested different versions of the model between the regression and probability tasks. Various tunings offered similar performance with some being slightly better at the probability task and some being slightly better at the regression. I opted for tunings that did well in the probability task to avoid false positives and negatives.

PR - AUC	t+1	t+3	t+5
Baseline	0.237	0.234	0.231
Model	0.346	0.344	0.366

Possible Improvements

- **Failure Mode:** Poor regression quality on rain rate in rainy settings.
- **Possible Solution:** Replace the Regression Head with a Dry Regression Head and a Rain Regression Head and gate them based on the result of the Probability Head.
 - e.g. $\text{reg}_{t+1m} = (1 - \text{sig}(\text{prob}_{t+1m})) * \text{Dry_Reg_Head} + \text{sig}(\text{prob}_{t+1m}) * \text{Rain_Reg_Head}$
- This approach was not as effective in the task data, but I think may improve with more heavy rain data to train on.